Andrew Hertzberg

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Education

The Pennsylvania State University, World Campus — Online
Master of Artificial Intelligence, in progress — *Aug. 2026 - Present*

The Pennsylvania State University — University Park, PA
B.S. in Computer Science, GPA: 3.50 — *May 2026*

Selected coursework: Machine Learning, Operating Systems, Systems Programming, Game Theory

Research and Experience

Next-Gen Innovators Fellow — Jul. 2026 - Present
Penn State Office of Technology Transfer — *University Park, PA*

- Selected for a yearlong research-commercialization fellowship with four-month rotations in university technology transfer, venture capital, and a technology startup.

Undergraduate Research Assistant — Jan. 2025 - Aug. 2025
The Pennsylvania State University — *University Park, PA*

- Developed and analyzed numerical ODE simulations of mRNA translation and ribosome dynamics, testing computational stability and behavior across parameter regimes.

Technical Research and Projects

LLM Message-Role Study — Aug. 2026 - Present
Python, PyTorch, Hugging Face Transformers, scikit-learn

- Extracted Qwen2.5-1.5B activations to compare representations of user text and tool-formatted text.
- Trained linear classifiers and investigated whether positional or dataset differences explained the results.

Systems Programming Projects
C, Pthreads, Linux, Make

- Built a custom allocator combining slab allocation for small objects with buddy allocation and coalescing for larger requests.
- Implemented FCFS, SRTF, and MLFQ scheduling with synchronized context-switch and I/O simulations in user space.

Graphics and Image Synthesis
C++, OpenGL, GLSL, OpenMP, Python, NumPy | CMPSC 458

- Extended a course framework with a recursive ray tracer supporting shadows, reflection, refraction, four-sample antialiasing, and an OpenMP render loop.
- Built heightmap terrain and spline-based roller-coaster geometry; implemented texture quilting with overlap matching and minimum-error seams.

Independent ML Systems Study

- Stanford CS336: Language Modeling from Scratch. Independent study of tokenization, Transformer components, optimization, training, and validation.
- How to Scale Your Model. Completed Chapters 1-6 and exercises on roofline analysis, TPU systems, sharded matrix multiplication, Transformer resource accounting, and training parallelism.

Presentation

Poster: "Numerical Study of a Traffic Flow Inspired by DNA Transcription Modeling," 2025 SIAM New York-New Jersey-Pennsylvania Section Conference, Penn State, Oct. 31-Nov. 2, 2025.

Technical Skills

Programming and ML: Python, C++, C, Java, PyTorch, Hugging Face Transformers, scikit-learn, NumPy
Tools and systems: Git, Linux, Pthreads, CMake, Make, OpenGL, GLSL